

4MA1_1F_ch2_equations_formulae_identities

Nguyen Khac Kiem PhD

1. 4MA1/1F_Winter_2025_Q27

1 point

2. 4MA1/1F_Winter_2025_Q15

1 point

3. 4MA1/1F_Winter_2025_Q4

1 point

4. 4MA1/1F_Summer_2025_Q23

1 point

5. 4MA1/1F_Summer_2025_Q14 1 point

6. 4MA1/1F_Summer_2025_Q4 1 point

7. 4MA1/1F_Summer_2024_Q25 1 point

(a) Simplify $(2p)^0$ where $p > 0$ (1)

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of n (1)

(c) Simplify fully $(5a^4c^2)^3$ (2)

A cylinder is placed on a table.

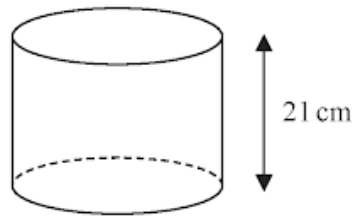


Diagram **NOT**
accurately drawn

The volume of the cylinder is 1575 cm^3

The force exerted by the cylinder on the table is 84 newtons.

$\text{pressure} = \frac{\text{force}}{\text{area}}$
--

Work out the pressure on the table due to the cylinder.

..... newtons/cm²

(3)

(a) Expand and simplify $(m + 5)(m - 8)$

(2)

(b) Solve $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

$n =$

(3)

(a) Simplify $10x - 7y - 6x + 4y$

(2)

$$T = 4d - 6e$$

(b) Work out the value of T when $d = 13$ and $e = 7$

$$T = \dots\dots\dots$$

(2)

(c) Solve $5p + 11 = 28$

$$p = \dots\dots\dots$$

(2)

Harold works in a factory.

His normal hourly rate of pay is £14

His overtime hourly rate of pay is £21

Harold is paid the normal hourly rate of pay for 35 hours in one week.

His total pay for this week is £679

Work out the number of hours of overtime he works in this week.

(4)

(a) Simplify $10p - 4p + 9p$

(1)

(b) Simplify $9 \times 4q$

(1)

(c) Solve $4r = 15$

$r = \dots\dots\dots$

(1)

(a) Factorise $y^2 - 2y - 48$

.....
(2)

(b) Write down the inequality shown on the number line



.....
(1)

(c) Solve the inequality $7w + 6 > 12w + 14$

.....
(3)

Solve $3(2 - 4x) = 5 - 8x$
Show clear algebraic working.

$x =$

3 marks

$$p = t - ac$$

$$t = 18$$

$$a = -3$$

$$c = 5$$

(a) Work out the value of p

$$p = \dots\dots\dots$$

(2)

(b) Make x the subject of the formula $d = 3x + 10$

$$\dots\dots\dots$$

(2)

(a) Simplify $5p \times 9k$

.....
(1)

(b) Simplify $3e + 2f + 8e - 7f$

.....
(2)

(c) Solve $2d + 7 = 16$

$d =$
(2)

↓ Solve the simultaneous equations

$$2x + 9y = 14.5$$

$$7x + 3y = 8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

(a) Simplify $(2c^4d^7)^3$

.....
(2)

(b) Find the value of $5y^0$ where $y > 0$

.....
(1)

(c) Factorise fully $16a^2b^3 + 20a^3b$

.....
(2)

(d) (i) Factorise $x^2 + 9x - 22$

.....
(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

.....
(1)

.....

.....

.....

.....

.....

(a) Simplify $p + p + p + p$

.....
(1)

(b) Simplify $5e + 6f + 7e - 2f$

.....
(2)

(c) Solve $13 - x = 7$

$x =$
(1)

(d) Solve $4y + 7 = 43$

$y =$
(2)

(a) Solve the inequality $5x - 7 \leq 2$

.....
(2)

(b) (i) Factorise $y^2 - 2y - 35$

.....
(2)

(ii) Hence, solve $y^2 - 2y - 35 = 0$

.....
(1)

(a) Simplify $a^7 \times a^4$

.....
(1)

(b) Simplify $w^{15} \div w^3$

.....
(1)

(c) Simplify $(8x^5y^3)^2$

.....
(2)

(d) Make t the subject of $c = t^3 - 8v$

.....
(2)

Alisa, Jena and Mikael each pick cucumbers.

Alisa picks C cucumbers.

Jena picks 5 fewer cucumbers than Alisa.

Mikael picks twice as many cucumbers as Alisa.

The total number of cucumbers picked by Alisa, Jena and Mikael is T

Find a formula for T in terms of C

Give your formula in its simplest form.

.....
3 marks

(a) Solve $5x = 30$

$x =$
(1)

(b) Solve $y - 7 = 12$

$y =$
(1)

(c) Simplify $h + h + h + h + h$

.....
(1)

(d) Simplify $5a + 7f - 2a + 4f$

.....
(2)

(a) Simplify $8 \times (4t)^0$

.....
(1)

$$x^6 \div x^{-5} = x^p$$

(b) Find the value of p

$p =$
(1)

(c) Simplify fully $(2k^2m^4)^3$

.....
(2)

(a) Expand and simplify $(n - 6)(n + 4)$

.....
(2)

(b) Solve $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

$x =$
(3)

(a) Simplify $12g - 8e - 5g + 6e$

.....
(2)

$$A = 3b - 5c$$

(b) Work out the value of A when $b = 12$ and $c = 4$

$A =$
(2)

(c) Solve $4p + 9 = 24$

$p =$
(2)

(a) Simplify $12a + 3a - 7a$

.....
(1)

(b) Simplify $8 \times 3b$

.....
(1)

(c) Solve $\frac{c}{3} = 9$

$c =$
(1)

(a) Simplify $(2x^3y^5)^4$

(2)

(b) (i) Factorise $x^2 + 5x - 36$

(2)

(ii) Hence, solve $x^2 + 5x - 36 = 0$

(1)

(a) Factorise fully $15y^4 + 20ny^3$

.....
(2)

(b) Solve $4 - 3x = \frac{5 - 8x}{4}$

Show clear algebraic working.

$x =$
(3)

(a) Expand $x(4 - x)$

.....
(1)

$$t = ab - c$$

$$a = 1.5 \quad b = 2.4 \quad c = -5.6$$

(b) Work out the value of t .

$$t = \text{.....}$$

(2)

(c) Make d the subject of $y = dx - e$

.....
(2)

(a) Simplify $10x + 4y + 3x - 6y$

.....
(2)

(b) Solve $2n + 5 = 16$

$n =$
(2)

(a) Factorise fully $25a^4c^7d + 45a^9c^3h$

.....
(2)

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x =$
(3)

(a) Expand and simplify $3x(2x + 3) - x(3x + 5)$

.....
(2)

(b) Make t the subject of the formula $p = at - d$

.....
(2)

Given that $\frac{w^5 \times w^n}{w^3} = w^{10}$

(c) work out the value of n .

$n =$
(2)

(a) Simplify $3 \times 10d$

.....
(1)

(b) Simplify $8e + e - 5e$

.....
(1)

(c) Solve $6g = 42$

$g =$
(1)

(d) Solve $24 = 10 + h$

$h =$
(1)

(a) Expand and simplify $(m - 8)(m + 5)$

.....
(2)

(b) Factorise fully $5y + 20y^2$

.....
(2)

(c) Simplify $(p^2 + 3)^0$

.....
(1)

The diagram shows a shape.

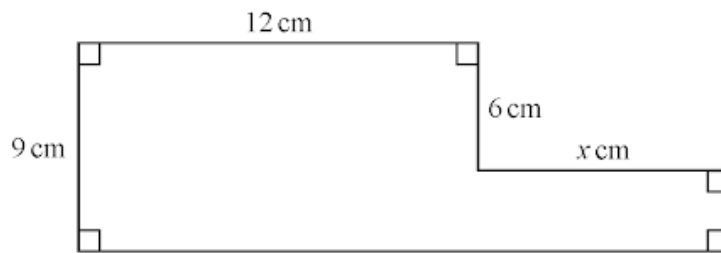


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

$x =$

4 marks

(a) Simplify $6m - 2k + 5m - k$

.....
(2)

$$P = 2a + 3b$$

(b) Work out the value of P when $a = 5$ and $b = 8$

$P =$
(2)

$$P = 2a + 3b$$

(c) Work out the value of a when $P = 16$ and $b = 20$

$a =$
(3)

- (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$$x = \dots\dots\dots$$

(3)

- (b) Factorise fully $16m^3g^3 + 24m^2g^5$

$$\dots\dots\dots$$

(2)

- (c) (i) Factorise $y^2 - 2y - 48$

$$\dots\dots\dots$$

(2)

- (ii) Hence, solve $y^2 - 2y - 48 = 0$

$$\dots\dots\dots$$

(1)

$$c = 4$$

$$d = 7$$

(a) Work out the value of $3c + 2d$

.....
(2)

$$p = -6$$

$$m = -2$$

(b) Work out the value of $2p^2 + 3m$

.....
(2)

There are 6 eggs in a small box of eggs.
There are 12 eggs in a large box of eggs.

Alex buys g small boxes of eggs and h large boxes of eggs.
He buys a total of T eggs.

(c) Write down a formula for T in terms of g and h .

.....
(3)

(a) Simplify $5c \times d$

.....
(1)

(b) Solve $x + 5 = 12$

$x =$
(1)

(c) Solve $9y = 36$

$y =$
(1)

(d) Simplify $8k + 5m - 2k + 6m$

.....
(2)

(e) Expand $4(3g + 1)$

.....
(1)

(a) Expand and simplify $(e + 3)(e - 5)$

(2)

(b) Solve $y = \frac{2y + 1}{5}$

Show clear algebraic working.

$y =$

(3)

(c) Solve $x^2 + 3x - 18 = 0$

Show your working clearly.

(3)

(Total for Question is 8 marks)

$$h = 6q - 2u$$

(a) Work out the value of h when $q = 3$ and $u = -5$

$$h = \quad (2)$$

$$B = 3m + 2p$$

(b) Work out the value of p when $B = 2$ and $m = 5$

$$p = \quad (2)$$

(a) Simplify $2e - 3f + 4e - 7f$

(2)

(b) Expand and simplify $5(4x + 3) - (3x - 1)$

(2)

(c) Factorise fully $4p + 6pq$

(2)

(a) Simplify $2e - 3f + 4e - 7f$

(2)

(b) Expand and simplify $5(4x + 3) - (3x - 1)$

(2)

(c) Factorise fully $4p + 6pq$

(2)

$$P = 2g + 3h$$

(a) Work out the value of P when $g = 7$ and $h = -4$

.....
(2)

(b) Simplify $e^9 \div e^5$

.....
(1)

(c) Simplify $(y^2)^8$

.....
(1)

(d) Expand and simplify $(x + 9)(x - 2)$

.....
(2)

(e) Factorise fully $16c^4p^2 + 20cp^3$

.....
(2)

(Total for Question 16 is 8 marks)

(a) Simplify $y^5 \times y^9$

.....
(1)

(b) Simplify $(2m^3)^4$

.....
(2)

(c) Solve $5(x + 3) = 3x - 4$
Show clear algebraic working.

$x =$
(3)

(d) (i) Factorise $x^2 + 2x - 24$

.....
(2)

(ii) Hence, solve $x^2 + 2x - 24 = 0$

.....
(1)

.....

.....

.....

.....

.....

(a) Simplify $6x + 8x - 3x$

.....
(1)

(b) Simplify $4e \times 5f$

.....
(1)

(c) Solve $8p = 24$

$p =$
(1)

(d) Solve $k - 4 = 13$

$k =$
(1)

(e) Simplify $10t + 4d - 3t + 2d$

.....
(2)

.....

.....

.....

.....

.....

